Assignment 1: Forced Alignment using Montreal Forced Aligner (MFA)

Objective
The objective of this assignment is to set up and execute a complete forced alignment pipeline using the Montreal Forced Aligner (MFA). The aim is to understand how automatic alignment works between speech audio and phonetic transcription at the word and phoneme level.

What is Forced Alignment
Forced alignment is the process of automatically matching an audio recording with its corresponding text transcription. Given the transcript, forced alignment determines when each word and phoneme begins and ends in the audio signal.

Dataset Description
The provided dataset consists of audio files in WAV format and corresponding transcript files in TXT format. Each transcript corresponds to one audio file and represents a single spoken utterance.

Environment Setup
The alignment pipeline was implemented using Montreal Forced Aligner version 3.3.9 in a Linux environment via Windows Subsystem for Linux (WSL). Praat software was used to inspect and visualize the alignment results.

Installation
MFA was installed using micromamba to maintain an isolated environment. The pre-trained English acoustic model and pronunciation dictionary were downloaded for alignment.

Data Preparation
Audio and transcript files were organized according to MFA requirements. Each audio file had a corresponding transcript with the same filename.

Dictionary Selection
The pre-trained English pronunciation dictionary *english_us_arpa* was used along with the corresponding acoustic model.

Forced Alignment
Forced alignment was performed using MFA, producing TextGrid files containing word-level and phoneme-level boundaries for each audio file.

Output Inspection and Analysis
The generated TextGrid files were inspected using Praat. The waveform was viewed alongside word and phoneme tiers to assess alignment quality. Word boundaries were generally accurate, with minor timing variations observed in fast speech regions.

Out-of-Vocabulary (OOV) Handling
No significant OOV words were observed in the dataset. In general, OOV issues can be resolved using MFA’s G2P models or by manually extending the pronunciation dictionary.

Conclusion
This assignment successfully demonstrates a complete forced alignment workflow using Montreal Forced Aligner. The results show effective alignment between audio and transcription at both word and phoneme levels.